Spectra: 5G Adaptive Beamforming and Channel Modeling Engine

Technical Systems Engineering Report

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December 2025

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1 Executive Summary

Spectra is a production-grade RF system simulation workstation designed to model, analyze, and visualize 5G physical-layer performance within a unified environment.

The platform integrates phased-array modeling, hybrid beamforming, massive MIMO SINR computation, OFDM simulation, and Monte Carlo analysis inside a real-time PyQt6-based GUI packaged as a standalone executable.

Unlike academic scripts, Spectra operates as an engineering workstation tool, bridging antenna physics, propagation modeling, and communication performance metrics in a deployable system.

$$\text{Antenna} \rightarrow \text{Channel} \rightarrow \text{Link Budget} \rightarrow \text{SINR} \rightarrow \text{Modulation}$$

2 System Architecture

Spectra follows a modular layered architecture separating simulation logic from presentation logic.

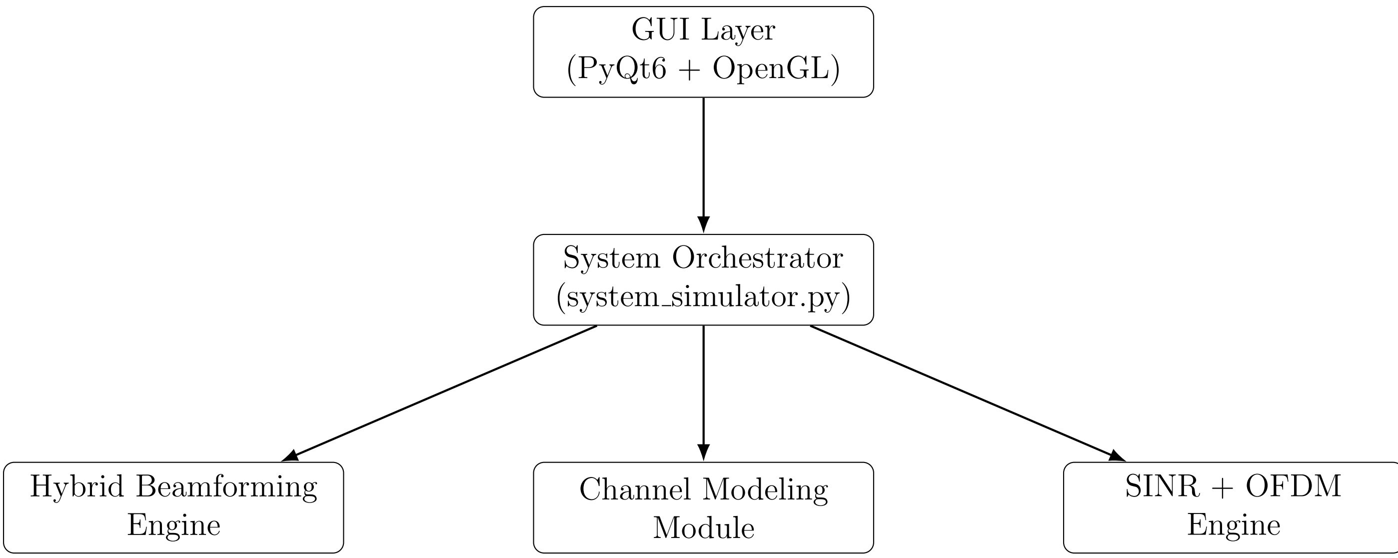

Figure 1: Spectra Layered Simulation Architecture

Core modules include:

- Channel modeling
- Hybrid analog-digital beamforming
- SINR computation
- OFDM modeling

- Link budget engine
- Power allocation controller

Threaded execution ensures simulation responsiveness while maintaining GUI stability.

3 Phased Array Modeling

Consider an $M \times N$ Uniform Rectangular Array (URA) with spacing d.

$$AF(\theta, \phi) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} w_{mn} e^{jkd(m \sin\theta \cos\phi + n \sin\theta \sin\phi)}$$

where:

$$k = \frac{2\pi}{\lambda}$$

Beam steering toward (θ_0, ϕ_0):

$$\psi_{mn} = -kd(m \sin\theta_0 \cos\phi_0 + n \sin\theta_0 \sin\phi_0)$$

Normalized radiation intensity:

$$|AF|_{norm} = \frac{|AF|}{\max(|AF|)}$$

4 Channel Modeling

Large Scale Fading:

$$PL(d) = PL_0 + 10n \log_{10}\left(\frac{d}{d_0}\right)$$

Small Scale Fading:

$$h \sim \mathcal{CN}(0, 1)$$

Monte Carlo sampling:

$$H^{(i)} \rightarrow SINR^{(i)}$$

This enables SINR distribution analysis under stochastic fading.

5 Link Budget and Noise Modeling

Friis transmission equation:

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2$$

Thermal noise:

$$N = kTB$$

Noise floor in dBm:

$$N_{dBm} = -174 + 10\log_{10}(B)$$

Signal-to-noise ratio:

$$SNR = \frac{P_r}{N}$$

Adaptive modulation is selected using predefined SNR thresholds.

6 Massive MIMO Precoding and SINR

Channel matrix:

$$H \in \mathbb{C}^{K\times M}$$

Zero-Forcing:

$$W = H^H (HH^H)^{-1}$$

MMSE:

$$W = H^H (HH^H + \sigma^2 I)^{-1}$$

User SINR:

$$SINR_k = \frac{|h_k^H w_k|^2}{\sum_{j\neq k} |h_k^H w_j|^2 + \sigma^2}$$

7 OFDM Modeling

IFFT operation:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}$$

Per-subcarrier SNR:

$$SNR_k = \frac{|H_k|^2 P_k}{N_0}$$

Subcarrier-level adaptive modulation enables realistic throughput estimation.

8 Advanced Features

- Monte Carlo SINR histogram and CDF visualization
- Adaptive modulation per subcarrier
- 3D radiation mesh rendering
- Beam weight magnitude and phase heatmaps
- Field deployment mode
- Executive PDF export

9 Software Engineering Design Decisions

Threading: Simulation executes in a QThread to prevent UI blocking.

Numerical Guardrails:

- Matrix regularization
- Noise floor clamping
- NaN protection
- Guarded normalization

Performance Optimization:

- NumPy vectorization
- Mesh caching
- Preallocated arrays
- Reduced redraw frequency

10 Validation Strategy

Validation includes:

- MATLAB cross-verification
- Analytical sanity checks
- Zero-noise identity channel testing
- Residual interference verification

$$\text{Physics First} \rightarrow \text{Numerical Verification} \rightarrow \text{Deployment}$$

11 Engineering Impact

Spectra demonstrates:

- Physical-layer RF understanding
- Hybrid beamforming implementation
- Massive MIMO signal processing
- Link budget computation expertise
- Numerical stability engineering
- Production-ready executable deployment

The project reflects system-level engineering maturity beyond academic simulations.

12 Future Extensions

- 3GPP CDL channel modeling
- Interference modeling
- Beam cut polar plots
- Performance sweep automation
- Configuration save/load
- Hardware-in-the-loop integration

13 Conclusion

Spectra is a unified RF system simulation platform integrating antenna physics, propagation modeling, massive MIMO beamforming, OFDM analysis, and Monte Carlo reliability evaluation within a deployable GUI framework.

The system bridges theory and implementation, demonstrating readiness for RF, analog, and physical-layer engineering roles.